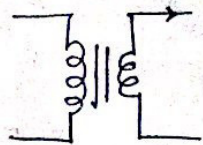


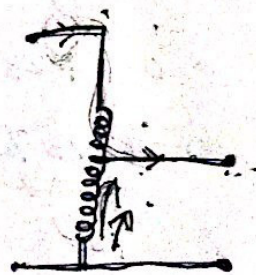
MODULE - V

Autotransformer.

- In a winding Xmer, windgs are electrically isolated and emf gets induced in secondary due to induction.
- It is possible to use one winding for transformer so that part of this winding is common to 1° & 2° .
- Such "a special type of Xmer having only one winding such that part of the winding is common to 1° & 2° is called Autotransformer.
- There is no electrical isolation b/w 1° & 2° windgs.
- It can provide maximum voltage transformation ratio of 2.
- In autoXmer, power is transferred from 1° to 2° as inductively as well as conductively.



2 winding Xmer.



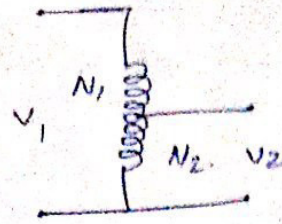
AutoXmer.

Autotransformer - Types.

a) step-down autoXmer.

It stepping down the input voltage applied to its 1° windgs.

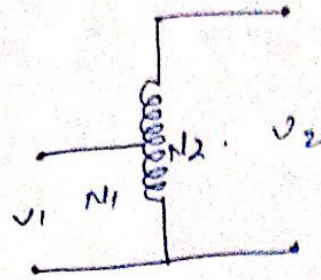
if 1° voltage $>$ 2° voltage



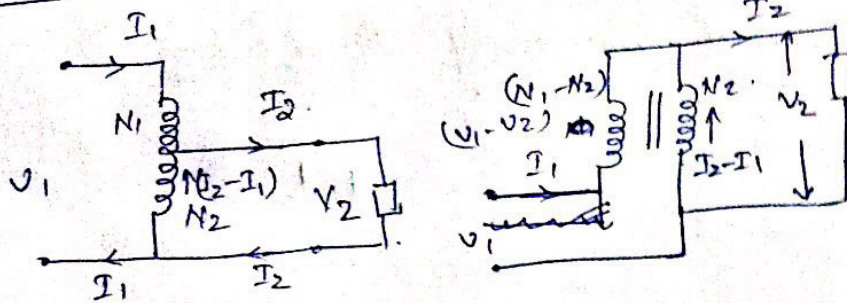
b) step-up autotrans.

It is stepping up the 1° voltage applied to its 1° windgs.

1° voltage $<$ 2° voltage.



Voltage Xmation Ratio.



$$\frac{V_1 - V_2}{V_2} = \frac{N_1 - N_2}{N_2}$$

$$V_1 N_2 - V_2 N_2 = V_2 N_1 - V_2 N_2$$

$$\frac{V_2}{V_1} = \frac{N_2}{N_1} = K$$

$$(V_1 - V_2) I_1 = V_2 (I_2 - I_1)$$

$$V_1 I_1 - V_2 I_1 = V_2 I_2 - V_2 I_1$$

$$\frac{V_2}{V_1} = \frac{I_1}{I_2} = K$$

$$\text{i/p power} = V_1 I_1$$

$$\text{o/p power} = V_2 I_2$$

power transferred inductively, $P_{\text{ind}} = V_2 (I_2 - I_1)$

$$= V_2 (I_2 - K I_1)$$

$$= K V_1 \left(\frac{I_1}{K} - I_1 \right)$$

$$= V_1 I_1 - K V_1 I_1$$

$$= (1-K) \text{ input power.}$$

power transferred conductively, $P_{\text{cond}} = P_{\text{out}} - P_{\text{ind}}$

$$= V_2 I_2 - (1-K) V_1 I_1$$

$$= K \times \text{input power.}$$

Copper Saving in Auto Xmer.

For any windg, the xsection of windg is proportional to current I and length of windg is $\propto$ to number of turns N . Hence weight of cu is $\propto$ to the product of N and I .

In 2-windg Xmer,

Weight of cu of 1^o $\propto N_1 I_1$

Weight of cu of 2^o $\propto N_2 I_2$

$\therefore$ Total weight of cu, $N_1 I_1 + N_2 I_2$

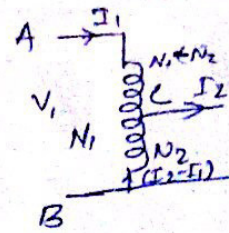
In case of auto xmer,

weight of cu of section AC

$$= (N_1 - N_2) I_1$$

weight of cu of section BC = $N_2 (I_2 - I_1)$

$\therefore$ Total weight of Cu, $W_{AT} = (N_1 - N_2) I_1 + N_2 (I_2 - I_1)$



$$\therefore \frac{W_{TW}}{W_{AT}} = \frac{N_1 I_1 + N_2 I_2}{N_1 I_1 - N_2 I_1 + N_2 I_2 - N_2 I_1}$$

$$= \frac{N_1 I_1 + N_2 I_2}{N_1 I_1 + N_2 I_2 - 2 N_2 I_1}$$

$$\frac{24 \times 25}{900} = \frac{100}{900}$$

$$K = \frac{N_2}{N_1} = \frac{I_1}{I_2}$$

$$\therefore \frac{W_{TW}}{W_{AT}} = \frac{N_1 I_1 + K N_1 \frac{I_1}{K}}{N_1 I_1 + K N_1 \frac{I_1}{K} - 2 \times K N_1 I_1}$$

$$= \frac{2 N_1 I_1}{2 N_1 I_1 - 2 K N_1 I_1} = \frac{1}{1 - K}$$

$$W_{AT} = (1 - K) W_{TW}$$

$$\therefore \text{Saving of Cu} = W_{TW} - W_{AT} = W_{TW} (1 - (1 - K))$$

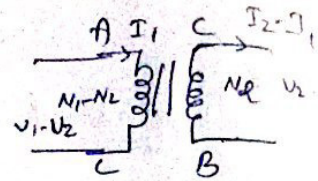
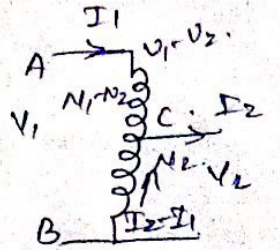
$$= \underline{K \times W_{TW}} \quad (\text{for step down xmer})$$

$$\ast \text{ Saving of Cu} = K \times \text{Weight of Cu in ordinary xmer.}$$

VA rating of Auto Xmer.

VA rating of Auto Xmer

$$(VA)_{AT} = V_1 I_1 = V_2 I_2$$



VA rating of 2 winding Xmer $= (V_1 - V_2) I_1 = (I_2 - I_1) V_2$
 $(VA)_{TW}$

$$= V_1 I_1 - V_2 I_1$$

$$= V_1 I_1 - K V_1 I_1$$

$$= \underline{\underline{V_1 I_1 (1 - K)}}$$

$$\frac{(VA)_{AT}}{(VA)_{TW}} = \frac{V_1 I_1}{V_1 I_1 (1 - K)}$$

$$\therefore \boxed{(VA)_{AT} = \frac{(VA)_{TW}}{(1 - K)}} \quad (\text{for stepdown AT})$$

Similarly, for step-up AT,

$$\boxed{(VA)_{AT} = \frac{K}{1 - K} \times (VA)_{TW}}$$

Adv. of AT.

- * Copper required is very less. & Cu losses are less.
- * η is higher as compared with 2 winding Xmer.
- * size & cost is less
- * VA rating is more.

Limitations

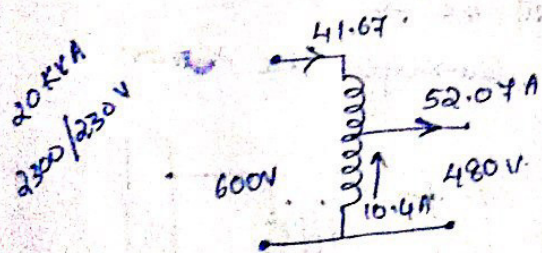
- * No electrical isolation b/w 1st & 2nd which is risky in case of high voltage levels.
- * Economical only where voltage ratio is less than 2.
- * If a section of winding common to P & 2nd is opened, full 1st voltage will be applied across 2nd, and resulting danger of accidents.

Appls.

- * Safe starting of loads like induction motor, syn. motors.
- * To give a small boost to a distribution cable to compensate for a voltage drop.

Q1/ A 480/120V, 5KVA 2 winding Xmer is to be used as an autotransformer to supply power at 480V from 600V source. Determine the KVA capacity of autotransformer.

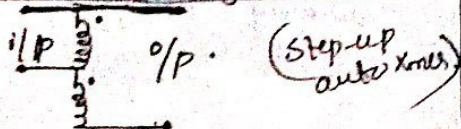
Ans: Current rating of 480V windgs = $\frac{5000}{480} = 10.4 \text{ A}$
 Current rating of 120V windgs = $\frac{5000}{120} = 41.67 \text{ A}$



$$\therefore \text{KVA rating of autotransformer} = \frac{52.07 \times 480}{1000} = \underline{\underline{25 \text{ KVA}}}$$

→ While converting 2-winding Xmer into autotransformer, 2 types of connections are possible.

1) Additive Polarity Connection



2) Subtractive Polarity Connection

